



Detecting fake review intentions in the review context: A multimodal deep learning approach

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ABSTRACT

The proliferation of fake reviews on the internet has had significant repercussions for both consumers and businesses. However, existing research predominantly employs a binary classification approach to ascertain review authenticity, often neglecting the rich multimodal context information and nuanced intentions embedded within them. To bridge this gap, our study introduces a novel task, Fake Review Intention Detection in Review Context (FRIDRC), which aims to detect fake review intentions by leveraging both textual and visual information, and constructs a dataset comprising both manually and AI-generated fake reviews. Additionally, we develop a predictive framework encompassing modules for multimodal representation and modality fusion. These modules, while independent, are synergistic and effectively tackle the challenge of discerning fake review intentions. Our framework demonstrates outstanding performance, achieving an average F1 score exceeding 0.97 and a Macro F1 score surpassing 0.96 in this task and outperforming advanced pre-trained models. This research not only presents an effective methodology for accurately identifying and addressing fake review intentions but also underscores the efficacy of leveraging multimodal review context information in fake review detection. The dataset and code implementation are publicly available for further research.

1. Introduction

Online reviews have emerged as a pivotal source of information, significantly influencing consumer purchasing decisions (Wang et al., 2023a) and affecting the sale strategies of companies (Wang et al., 2023b). These reviews not only offer product insights but also encapsulate shared consumer experiences, thus serving as a critical element in shaping brand image and driving sales. However, the proliferation of online reviews has given rise to the challenge of fake reviews (Patel and Patel, 2018; Wu et al., 2020a), posing considerable risks to both consumers and businesses (Lappas et al., 2016; Mohawesh et al., 2021). Recent instances underscore the severity of this issue; Tripadvisor, one of the largest online travel agencies, disclosed the detection of 1.3 million fake reviews in 2022 (Tripadvisor, 2023), and a report by the UK government in 2023 estimated that fake reviews inflict economic losses of £500 million to £3.1 billion annually on UK consumers (Alma Economics, 2023).

This escalating concern over fake reviews has catalyzed extensive academic investigation. Research efforts have expanded beyond identifying fake reviewers (Wang et al., 2012; Liu and Pang, 2018; Cao et al.,

2019; Zhao et al., 2023), and groups (Wang et al., 2016; Zhang et al., 2018; Wang et al., 2018; ZHA, 2020) to encompass the detection of fake reviews themselves (Mohawesh et al., 2021), with a particular focus on assessing the authenticity of review content. Predominantly, such endeavors treat fake review detection as a binary classification task, relying solely on textual content (Mukherjee et al., 2021; Fontanarava et al., 2017; Elmogly et al., 2021; Hajek et al., 2020). This approach, however, is reductive and overlooks crucial aspects: it neglects the multifaceted intentions behind the generation of fake reviews (Zaman et al., 2023) and, by concentrating exclusively on review text, disregards the multimodal context of reviews, including elements such as visual representations (e.g., images or videos) that provide critical context about the subjects being reviewed. Acknowledging the importance of both semantic and contextual information from the review content and multimodal sources, especially when combining text and images, has proven beneficial in related domains such as review usefulness inference (Liu et al., 2021; Nguyen et al., 2023) and sentiment analysis (Wu et al., 2024).

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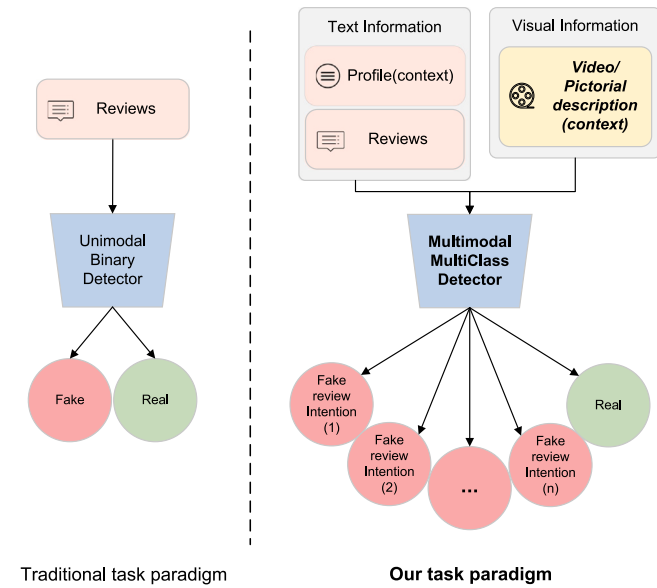


Fig. 1. Task difference between FRIDRC and traditional fake review detection. Our task utilizes not only the review text but also integrates visual cues from video frames(images) associated with the reviews. These visual elements are crucial in setting the context for the text and enriching the model's ability to discern the underlying intentions behind the reviews.

To bridge the identified research gap, this study endeavors to elucidate the nuanced intentions behind fake reviews and to detect these intentions within the review context. We coin this novel task as *Fake Review Intention Detection in Review Context (FRIDRC)* which moves beyond the traditional binary classification paradigm by incorporating a richer, multimodal perspective to address the complexities of fake review detection. Fig. 1 illustrates the difference between the proposed novel task and the traditional fake review detection task.

Achieving this goal necessitates exploring the primary intentions underpinning fake reviews. To facilitate context-aware detection, it is imperative to develop a dataset that encompasses diverse intentions behind fake reviews and relevant review context features. To the best of our knowledge, existing studies lack such a comprehensive dataset. Therefore, we articulate our initial research questions as follows:

- **RQ 1.1:** What are the primary intentions behind fake reviews?
- **RQ 1.2:** How can a dataset be constructed to reflect these intentions and associated review contexts?

Following the dataset's preparation, devising an approach to the review context may encompass multiple modalities, such as textual descriptions and visual illustrations found in product listings on e-commerce platforms. Thus, the task at hand inherently aligns with multimodal classification challenges (Kiela et al., 2018; Wang et al., 2020). Consequently, our investigation extends to the formulation of a multimodal deep learning approach tailored for the FRIDRC task, along with an evaluation of its efficacy. This leads to the formulation of our subsequent research questions:

- **RQ 2.1:** How can a multimodal deep learning methodology be designed for the proposed task?
- **RQ 2.2:** How does multimodal deep learning perform on the FRIDRC task, and what is the significance of context features in this domain?

This paper is structured as follows: Section 2 provides a review of the literature on fake review detection and multimodal classification. Section 3 details our exploration of fake review intentions and the development of a supportive dataset for the FRIDRC task. Section 4 outlines our proposed methodology for tackling the FRIDRC task, followed

by Section 5, which presents our experimental evaluation. Section 6 further discusses our key findings, implications, limitations and directions for future research. Finally, Section 7 concludes this study.

2. Related work

This research investigates the novel task of utilizing multimodal deep learning for fake review detection, a field where most existing studies focus solely on textual analysis. Given that fake reviews are a form of online misinformation (Salminen et al., 2022), insights can be drawn from multimodal deep learning approaches used in broader misinformation detection. This section outlines the landscape of deep learning in fake review detection and the application of multimodal techniques in identifying online misinformation.

2.1. Deep learning-based fake review detection

Fake review detection is primarily a text classification task. Early methods leveraged traditional word embedding techniques, with studies like Hajek et al. (2020) demonstrating the effectiveness of convolutional neural network (CNN) and deep feed forward neural network (DFFNN) models over advanced methods across multiple datasets. Similarly, Al-Adhaileh and Alsaade (2022) and Baishya et al. (2021) highlighted the superiority of deep learning models, such as bidirectional long short-term memory (BiLSTM) and a combination of Convolutional 2D, BiLSTM, and Multichannel CNN1D with LSTM, in identifying fake reviews on platforms like Yelp and Amazon. In addition to supervised learning, unsupervised approaches have shown promise. For instance, Neisari et al. (2021) successfully applied a combination of Self-Organizing Map (SOM) and CNN to classify reviews without labeled data, achieving high accuracy in both single-domain and multi-domain environments. Then, the advent of pre-trained language models has further advanced the field. Studies utilizing models like RoBERTa and BERT have fine-tuned these architectures on specific datasets to detect fake reviews with high accuracy (Salminen et al., 2022; Catelli et al., 2022). Gupta et al. (2022) explored multiple Transformer-based models, concluding that RoBERTa-based classifiers excel in this task. In addition, feature engineering plays an important role in identifying fake reviews. Satia Budhi et al. (2021) proposed 133 unique features derived from a combination of content-based and behavior-based feature extraction approaches to be used with machine learning classifiers for detecting fake reviews.

Despite these advancements, current research predominantly focuses on textual content, overlooking the potential of visual and semantic cues. The exploration of multimodal features in fake review detection remains an area ripe for investigation.

2.2. Multimodal deep learning-based online disinformation detection

Fake reviews are a form of disinformation, yet there is limited research on detecting them using multimodal deep learning methods. In this paper, we review the advancements in disinformation detection. Recent research in disinformation detection has made significant strides using multimodal deep learning, which leverages both text and visual information. Meel and Vishwakarma (2021) achieved a notable 95.9% accuracy in fake news detection through Hierarchical Attention Networks, image captioning, and forensic analysis. Kumari and Ekbal (2021) and Palani et al. (2022) demonstrated the effectiveness of multimodal approaches in enhancing detection accuracy, with methods reaching an F1 score of 0.83 and classification accuracies above 92%, respectively. Singh et al. (2023) and Hua et al. (2023) further validated the potential of multimodal deep learning by achieving high accuracy and F1 scores in detecting fake news. The Multimodal Variational Autoencoder (MVAE) proposed by Khattar et al. (2019) and the multimodal data fusion techniques by Choi and Ko (2022) outperformed existing models, highlighting the advantage of

integrating diverse data types. Additionally, multimodal learning has been applied beyond news to social media content, with models like those developed by Fontanarava et al. (2017) and Qi et al. (2023) effectively detecting misinformation through advanced feature selection and leveraging social context.

These advancements underscore the robustness of multimodal deep learning in identifying online disinformation, setting a foundation for further exploration in integrating varied content types for enhanced detection accuracy.

2.3. Large language model-based data creation

Large language model (LLM)-based data creation leverages LLMs' text generation capabilities to efficiently produce large volumes of synthetic data, offering a promising solution across domains where annotations are typically difficult or expensive to obtain. In dialogue summarization, Chintagunta et al. (2021) employed GPT-3 to generate synthetic medical dialogue summaries, demonstrating that training models on a combination of synthetic and human-labeled data could scale performance from a small human-labeled dataset to levels comparable to models trained with 30 times the amount of human-labeled data. Similarly, Wan et al. (2022) fine-tuned language models on limited samples to generate synthetic dialogues in target domains. In few-shot classification, Møller et al. (2023) provided examples with corresponding labels and instructed LLMs to generate similar labeled examples, leading to improved downstream performance. In intent detection, Lin et al. (2023) used OPT-66B to generate synthetic examples based on the training set, while Sahu et al. (2022) prompted GPT-3 to generate training data that boosted performance in intent classification. Additionally, Salminen et al. (2022) used GPT-2 to augment a dataset of fake reviews, demonstrating the model's ability to generate convincing data that could deceive human evaluators. These examples underscore the versatility of LLMs in synthetic data generation, providing a scalable and cost-effective approach to addressing data insufficiency across diverse applications.

3. Dataset

Numerous datasets are available for fake review detection, including Yelp CHI (Mukherjee et al., 2013), Yelp NYC (Rayana and Akoglu, 2015), Chicago Hotels (Ott et al., 2011), and Epperly (Li et al., 2011). However, these datasets do not meet our study's needs because they lack data on the diverse intentions behind fake reviews and multimodal review contexts. Consequently, it is essential to develop a new dataset tailored to the specific requirements of our proposed Fake Review Intention Detection in Review Context (FRIDRC) task.

Constructing a dataset for fake review detection typically involves collecting a large corpus of online reviews and employing annotators to determine the authenticity of these reviews. A significant limitation of this method is the rarity of fake reviews; for example, only 4.37% of Yelp reviews are fraudulent (Tripadvisor, 2023). This scarcity makes it difficult to obtain a sufficient number of fake reviews that represent specific intentions.

To overcome this challenge and address RQ 3.2, we pursued a generative strategy to assemble a dataset focused on the intentions of fake reviews. Initially, we collected high-quality online reviews as baseline negative samples. Then, drawing on the extensive literature on fake review intentions, we identified the most prevalent intentions behind such reviews. Annotators were then tasked with crafting fake reviews that emulate the style of authentic reviews, reflecting these identified intentions. To further enrich our dataset, we intend to leverage advanced generative language models to produce human-like fake reviews.

Table 1
Intentions of fake reviews.

Intentions	Description and supporting literature
Incentivized Exaggeration (IE)	Merchants induce reviewers to overly praise products or services through rewards to attract more consumers.
Competitive Denigration (CD)	Competitors invite reviewers to deliberately disparage the quality of competing products to diminish their market appeal
Traffic attraction (TA)	Reviewers introduce unrelated topics to divert attention or transfer interest to related products or services, such as linking car insurance in automotive reviews.
Irrelevant Commentary (IC)	The content of the review has no connection to the subject of the review, often posted in irrelevant review sections for a certain agenda.

The intentions behind fake reviews can be categorized into four types which reflect the various motivations behind fake reviews, providing a more analytical perspective for the FRIDRC task while addressing RQ 1.1.

3.1. Intentions to make fake reviews

Our review of the literature identifies financial incentives as the primary motivator for review manipulation (Wu et al., 2020a; Zhang et al., 2022; Petrescu et al., 2018). Notably, Anderson and Magruder (2012) and Chevalier and Mayzlin (2006) found that positive reviews significantly boost a merchant's sales, prompting some to solicit false positive reviews for economic gain. Additionally, competition is another key driver; some merchants generate negative reviews to undermine competitors and damage their reputations (Crawford et al., 2015; Lappas, 2012). Studies, such as Lappas et al. (2016), have demonstrated that just 50 fabricated reviews can significantly enhance a business's visibility, whereas negative reviews can severely impact the sales of less established brands (Ho-Dac et al., 2013). We categorize these intentions as “**Incentivized Exaggeration (IE)**” and “**Competitive Denigration (CD)**”.

Another prevalent form of fake reviews includes those that serve as advertising or contain irrelevant content (Jindal and Liu, 2008; Rădulescu et al., 2014; Alsaleh et al., 2015; Zhang et al., 2017; Li et al., 2019; Huang, 2022). These reviews typically lack detailed discussion about the product's quality or usefulness, and instead use vague, generic, or unrelated language. We classify these under two categories: “**Traffic Attraction (TA)**”, which leverages online reviews to divert readers to other products or services related to the reviewed item, such as advertorials in online reviews (Chang et al., 2018). Another is “**Irrelevant Commentary (IC)**”, encompassing cases where completely unrelated information is pushed, or reviews are copied from other products (Li et al., 2019; Huang, 2022). Details about each kind of fake review intention are listed in Table 1.

3.2. Dataset creation

The dataset creation process, depicted in Fig. 2, begins with the collection of legitimate online reviews. This is followed by the manual generation of fake reviews, which are designed to reflect various underlying intentions. The final step in the dataset development involves augmenting the volume of fake reviews using artificial intelligence, specifically generative language models.

3.2.1. Review data collection

Given that our data annotators are predominantly Chinese university students, we have tailored our dataset construction to reflect the Chinese context. We selected Douyin (douyin.com), a prominent Chinese short video platform on social media, as our primary data source.

To ensure the robustness and representativeness of our fake review dataset, we selected samples from ten representative subcategories

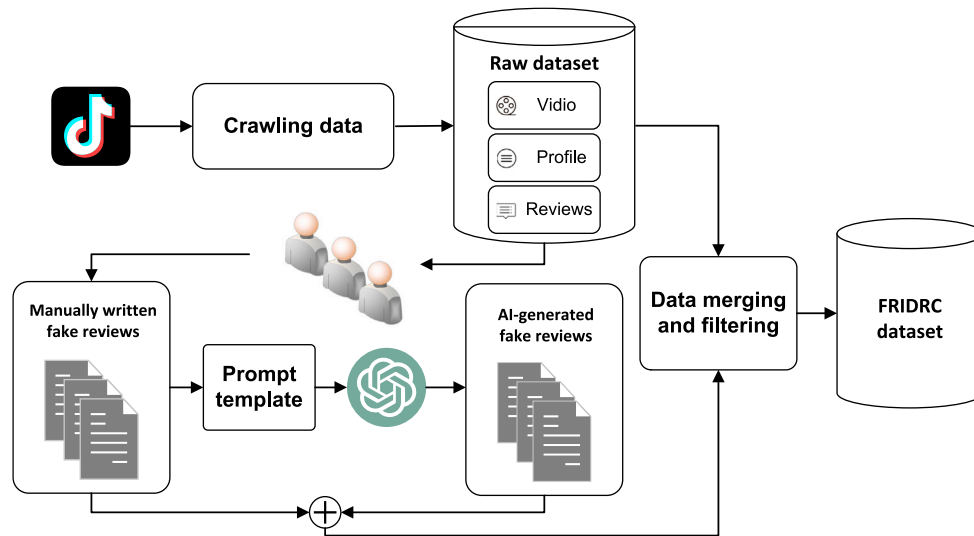


Fig. 2. Dataset construction workflow. The process starts by collecting genuine reviews and contextual data from Douyin. Fake reviews are then manually created by annotators based on predefined intentions (e.g., incentivized exaggeration, competitive denigration). Additionally, AI-generated fake reviews are produced using a generative language model with few-shot learning. After quality filtering, both human and AI-generated reviews are combined to form the final dataset, which is designed for supporting fake review intention detection in a multimodal context.

of online products and services. These categories include *Restaurants*, *Health*, *Hotels*, *Professional Services*, *Educational Institutions*, *Automobile*, *Clothing*, *Beauty*, *Electronics*, and *Food and Beverage Products*. They were chosen due to their high levels of consumer demand and interaction, which are crucial for creating a diverse and comprehensive dataset. For each category, we focused on gathering reviews and contextual data (including videos and descriptions) from the most popular items. The selection process was guided by the popularity of each item, determined by the number of likes its associated video received, starting with the most liked items in each category and continuing in descending order. The process of extracting images from videos was accomplished through a frame selection technique, specifically utilizing a frame difference algorithm which works by calculating the pixel-level differences between consecutive frames. Instead of selecting the first frame or keeping all frames, we applied this algorithm to identify the most representative frame for each video.

Following data collection, we enlisted ten college students who are active users of the Douyin platform to serve as reviewers. Each volunteer was assigned to generate fake reviews within their area of expertise, aligning with specific intents. After a thorough vetting process, we compiled a total of 400 crafted fake reviews, distributed evenly with 40 reviews per video category and 100 comments per intent type. These reviews were then utilized to create few-shot learning prompts, further enhancing the dataset's utility for model training.

3.2.2. Fake review augmentation

The advent of advanced LLMs like ChatGPT has significantly exacerbated the challenge of fake reviews by enabling the generation of highly deceptive content at an unprecedented scale (Knoedler et al., 2024; Ho et al., 2024). These models possess remarkable natural language generation capabilities, producing reviews that are sophisticated and indistinguishable from those written by humans. This advancement poses a substantial threat to the integrity of online review ecosystems and underscores the necessity for innovative detection methods. Our study addresses this issue by not confining our dataset to solely human-generated content. Following the approach by Salminen et al. (2022), we incorporate AI-generated fake reviews. This strategy serves dual purposes: it broadens the spectrum of fake reviews in our dataset and increases the complexity of the detection task.

We employed the few-shot learning methodology (Brown et al., 2020), a technique that improves a model's task performance with

Table 2
Summary of data metrics by category.

Class	Subclass	Real	IE	CD	TA	IC
Service	Restaurants	1759	449	464	449	437
	Education	2964	462	442	452	436
	Hotel	1296	431	433	449	452
	Professional services	2417	449	436	444	456
	Health	743	460	463	439	435
Merchandise	Automobile	1820	448	451	438	450
	Clothing	1371	445	449	442	441
	Cosmetics	1446	445	441	456	432
	Electronics	2414	447	454	463	440
	Food and beverage	1638	450	481	405	452

minimal prior data exposure. For generating each AI fake review, we used 10 selected artificial fake reviews as prompts. These prompts were integrated into templates that guided the model in creating review content. The procedure for generating a fake review for a video is depicted in Fig. 3. We then merged these AI-generated reviews with genuine and manually crafted fake reviews. To improve the dataset's quality, we implemented quality control measures such as excluding overly brief reviews. Table 2 delineates the dataset's structure.

4. Multimodal framework for fake review intent detection

In this study, FRIDRC is approached as a multimodal classification task, a domain that has been extensively explored. Instead of devising a new multimodal algorithm for FRIDRC, our objective is to develop a framework that integrates state-of-the-art multimodal classification techniques. Drawing inspiration from Sleeman et al. (2022), we have constructed an early-fusion multimodal framework for detecting fake review intent using decoupled modules.

As illustrated in Fig. 4, the framework comprises three main modules: Text feature representation, image feature representation, and feature fusion. Each module operates independently yet synergistically. Specifically, the text and image feature representation modules allow for the selection of various feature encoders to optimize performance; concurrently, the framework offers flexible adjustment of the feature fusion strategy. This adaptability helps in effectively capturing interactions between different modalities and minimizing redundant information during model training. The subsequent sections will provide detailed descriptions of each module in the framework.

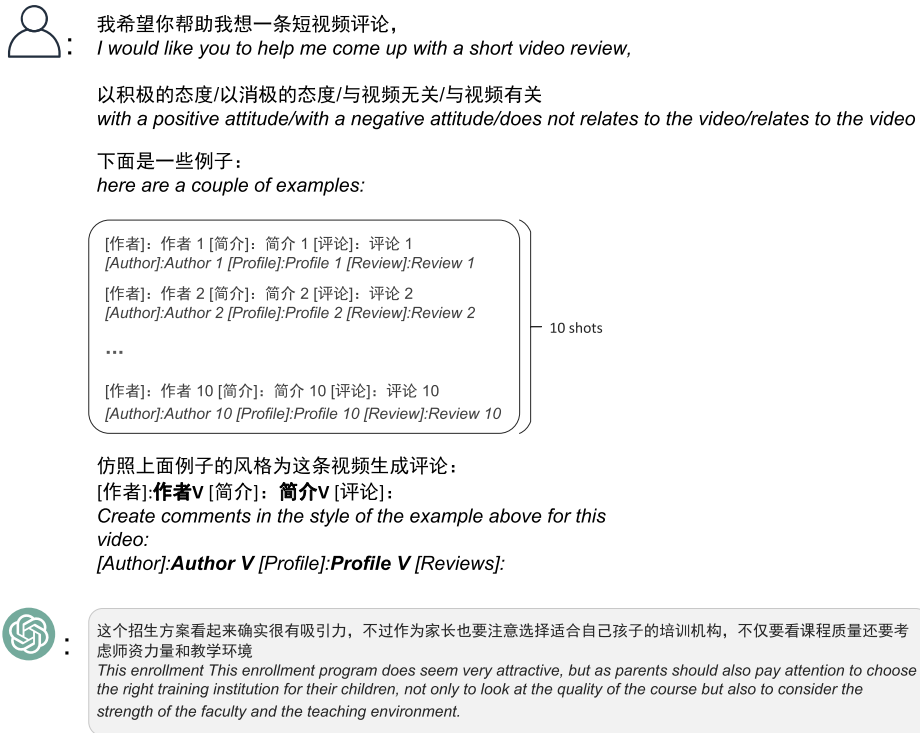


Fig. 3. Prompt template for generating fake online reviews. “Author V” and “Profile V” represent the author and profile of video V, respectively. The format “[Author]:Author 1 [Profile]:Profile 1 [Review]:Review 1” denotes a manually written fake review along with the corresponding author and profile of the video.

4.1. Motivations

Most state-of-the-art pre-trained multimodal models rely on well-aligned image-text pairs to learn effective cross-modal representations. However, in the task of fake review detection, the relationship between textual and visual data is often complex, making it challenging to handle alignments. This misalignment between pretraining models’ prior knowledge and the specific challenges of the fake review detection task may contribute to the suboptimal performance observed in our experiments (see Table 4). As a result, we identify the need to develop a tailored multimodal framework that better addresses these complexities.

Another key aspect of our approach is the adoption of a flexible, modular framework that allows each component — such as text representation, image representation, and feature fusion — to be independently optimized and upgraded. This modularity enhances adaptability to new advancements, ensuring that the framework remains competitive in the evolving domain of fake review detection over the long term.

Besides, a key aspect of multimodal classification, especially in tasks such as the detection of fake reviews, lies in exactly how effectively the various modalities can be fused to form one unified representation. We therefore explore five different strategies for the fusion of modalities, so as to investigate how different approaches toward fusion affect the performance of the overall model and provide valuable insights into the most effective methods that could integrate textual and visual data within this context.

4.2. Modality representation

This subsection outlines the methodologies and technologies employed to represent text and image data. Our framework utilizes typical and advanced models tailored for the specific characteristics of each modality, ensuring that the data is processed in a manner that maximizes its utility for FRIDRC task.

4.2.1. Text feature representation

Given that all review texts in our dataset are in Chinese, we utilize two pretrained language models specifically developed for the Chinese language: Chinese-BERT and Chinese-RoBERTa.

Chinese-BERT. In this study, we employ the Chinese-BERT-wwm-ext (Cui et al., 2020) (henceforth Chinese-BERT) as a representation model for analyzing review texts. Chinese-BERT leverages a substantial corpus, including 5.4 billion words from various Chinese sources. The BERT architecture (Devlin et al., 2019) involves tokenizing the text into a sequence, which is then embedded and enriched with position-specific data. The resultant embeddings undergo transformation by a Transformer encoder that consists of 12 layers employing multi-head self-attention and feed-forward networks, ultimately producing a 768-dimensional feature vector of the text.

Chinese-RoBERTa. Alongside Chinese-BERT, we utilize the Chinese-RoBERTa-wwm-ext (Cui et al., 2021) (referred to as Chinese-RoBERTa), which integrates the Whole Word Masking technique with the foundational RoBERTa architecture to enhance natural language processing outcomes. Distinct from BERT, RoBERTa (Liu et al., 2019) omits next-next-sentence prediction and dynamic masking, operates with longer sequence lengths from the outset, and extends the training duration to one million steps. The RoBERTa model processes the review texts similarly by tokenizing, embedding, and position encoding the inputs before they are transformed by the same Transformer encoder setup, resulting in a 768-dimensional output vector comparable to that of BERT.

4.2.2. Image feature representation

In addition to text, image data provides essential visual cues that are instrumental in understanding the context of online reviews. For our image feature representation, we have adopted two distinct approaches: the widely-used ResNet-50, known for its efficiency and effectiveness, and the more advanced Vision Transformer (ViT).

ResNet-50. ResNet-50 (He et al., 2016) is utilized to extract high-level features from input images in our analysis. Initial processing standardizes the input image to 224×224 pixels with three channels.

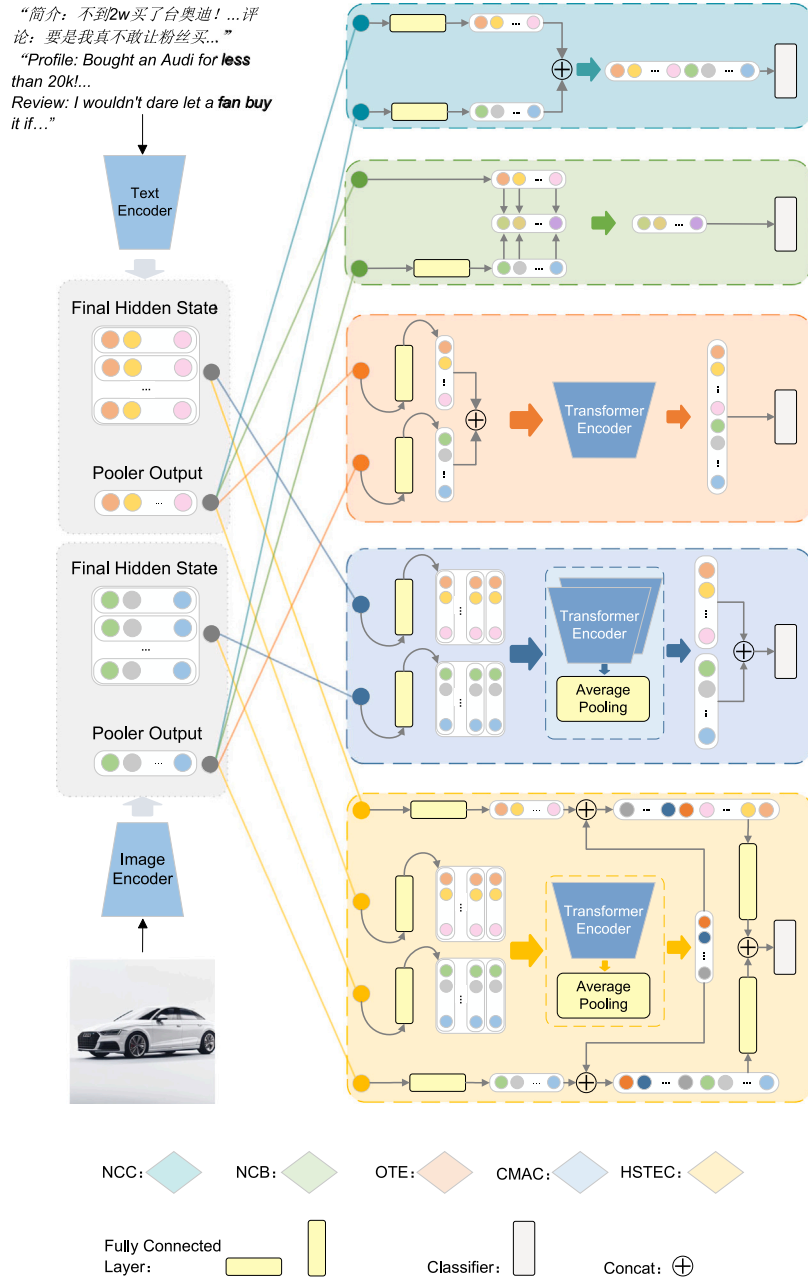


Fig. 4. Multimodal framework for fake review intent detection. The five fusion methods are differentiated by distinct colors, with the colored lines indicating the input flow through the respective modules. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

The image is then subjected to a 7×7 convolution and a 2×2 max pooling, resulting in intermediate feature representations. The image data passes through four sequential modules of residual blocks that refine features while adjusting the resolution and channel depth. Ultimately, the process concludes with average pooling and flattening, yielding a 2048-dimensional feature vector. This sequential enhancement through residual learning helps in capturing intricate details necessary for robust image analysis.

Vision Transformer (ViT). The Vision Transformer (ViT) (Wu et al., 2020b) serves as another image representation method, which departs from traditional convolutional approaches by treating images as sequences of patches. Each image is resized to 224×224 pixels, and segmented into patches, and each patch is embedded into a higher-dimensional space. ViT introduces a class embedding to these sequences along with positional encodings to retain spatial hierarchy. The sequence of embeddings is processed through multiple layers of

a Transformer encoder, consisting of multi-headed self-attention and MLP blocks, enhanced with LayerNorm and residual connections. The output from the Transformer's final layer, after normalization, forms a 768-dimensional vector representing the image's features. This method leverages the Transformer's ability to capture global dependencies across image patches, enhancing the model's interpretative capability.

4.3. Modality fusion strategies

This subsection introduces five modality fusion strategies implemented within our framework, each employing unique mechanisms to merge textual and visual information. The architectural designs of these models are depicted in Fig. 4. The subsequent paragraphs detail various strategies for alignment, projection, and interaction between modalities.

Naive concat. The Naive Concat (NCC) (Iandola et al., 2014) handles pooler outputs from text and image modalities. For details on its architecture. Initially, the NCC model projects the outputs of text and image encoders into the desired dimensional space, represented as V_t' and V_i' . Subsequently, these features are concatenated and fed into a linear classifier C to obtain the classification logits:

$$\text{Logits} = C(\text{Concat}(V_t', V_i')). \quad (1)$$

Naive combine model. The Naive Combine (NCB) (He et al., 2016), also processes text and image pooler outputs. Unlike the NCC, NCB aligns and sums these outputs element-wise, generating the integrated features E' . These are then evaluated by a linear classifier C to produce the classification logits:

$$\text{Logits} = C(E'). \quad (2)$$

Output transformer encoder model. The Output Transformer Encoder (de Toledo and Marcacini, 2022) processes the pooler outputs V_t and V_i , projecting them into V_t' and V_i' for compatibility. These projections are concatenated:

$$X' = \text{Concat}(V_t', V_i'). \quad (3)$$

This concatenated output X' is then passed through a multi-head self-attention mechanism, resulting in A_{output} . This output is used to compute the classification logits:

$$\text{Logits} = C(A_{\text{output}}). \quad (4)$$

Cross-modal attention combine model. The Cross-Modal Attention Combine model (CMAC) (Zheng, 2022) utilizes the last hidden states H_t' and H_i' and incorporates two multi-head attention processes for text-to-image and image-to-text interactions, defined as: $\text{Multihead}_{\text{text-image}}$ and $\text{Multihead}_{\text{image-text}}$.

The outputs from these interactions, $A_{\text{text-image}}$ and $A_{\text{image-text}}$, are averaged and concatenated, serving as the input for the classifier to determine the logits:

$$\text{Logits} = C(\text{Concat}(\text{Mean}(A_{\text{text-image}}), \text{Mean}(A_{\text{image-text}}))). \quad (5)$$

The hidden state transformer encoder model. The Hidden State Transformer Encoder (HSTEC) (Zheng, 2022) integrates pooled outputs and the last hidden states from text and image encoders, processed through a multi-head attention to obtain A_{output} . This output, together with the transformed pooled outputs, feeds into fully connected layers to extract text and image features:

$$F_t = \text{FC}_{\text{text}}(\text{Concat}(A_{\text{output}}, V_t')) \quad \text{and} \quad F_i = \text{FC}_{\text{image}}(\text{Concat}(A_{\text{output}}, V_i')). \quad (6)$$

Finally, the combined features F_t and F_i are concatenated and used to compute the classification logits:

$$\text{Logits} = C(\text{Concat}(F_t, F_i)). \quad (7)$$

5. Experiments

5.1. Experimental setup

5.1.1. Dataset

The dataset employed in this study was constructed as described in Section 3. To balance the dataset, the number of samples containing real reviews was made equal to the number of fake reviews, with an approximately equal distribution across different types of fake intentions. The data was then split into training and test sets in a 4:1 ratio, respectively. The distribution of the dataset is detailed in Table 3.

Table 3

Class distribution.

	R	IE	CD	TA	IC	Sum
Train set	14,294	3611	3589	3549	3545	28,588
Test set	3574	903	897	888	886	7148

5.1.2. Experimental setting

The programming language used in this experiment is Python, version 3.10.9, and the deep learning framework is PyTorch, version 2.0.0. The hardware environment used in this experiment includes a Supermicro GPU server, with dual Xeon 8276L processors, 1TB of memory, and 10 NVIDIA RTX 3090 24G graphics cards.

We assessed model performance using macro precision, macro recall, and macro F1 score. The F1 score was chosen as a key evaluation metric because it provides a balanced measure of precision and recall. Additionally, average precision, average recall, and average F1 score were calculated to provide further insights into model accuracy across various classes.

5.2. Baseline

This section describes the baseline model used for comparison of experimental results.

- Taiyi-CLIP (Zhang et al., 2023) is a Chinese CLIP model that tunes the Chinese-RoBERTa-wwm text encoder on 123 million graphic pairs, and the image encoder is from ViT-L-14 in Open-Clip. It follows the experimental setup of CLIP to obtain powerful visual-language intelligence.
- Qwen-VL (Qwen Large Vision Language Model) (Bai et al., 2023) is the multimodal version of the large model series, Qwen (abbr. Tongyi Qianwen), proposed by Alibaba Cloud. Qwen-VL outperforms current SOTA generalist models on multiple VL tasks (at the time of model release) and has a more comprehensive coverage in terms of capability range.

5.3. Main results

The initial experiment was designed to assess the efficacy of our proposed multimodal framework in identifying fake review intentions. We compared the performance across various input configurations: review text only, review text combined with images, and review text with both profile information and images. Additionally, we evaluated two pre-trained multimodal models. The results are detailed in Table 4, and the setting that achieves best performance is shown in Table 5. Notably, the configuration combining review text, profile information, and image information achieved the highest average and macro F1 scores, at 0.97 and 0.96, respectively, surpassing other input setups and baseline models. The combination of review text and image information also demonstrated robust performance, with scores exceeding 0.95, higher than when using only review text. Interestingly, despite having more parameters, the baseline models underperformed compared to the RoBERTa model, which utilized solely review text. This underperformance might be attributed to the inappropriateness of the large unified modality model for the FRIDRC task or to parameter overfitting.

These findings preliminarily address RQ 2.2 that integrating multimodal information from review contexts can improve the detection performance of fake review intentions. However, it should be noted that the data presented in Table 4 only represent the best-performing experimental settings for each input combination and fusion method. The robustness of the models across different inputs will be discussed further in subsequent sections.

Table 4
Main results.

Input	Model		Fuse type	Average			Macro		
	Text encoder	Image encoder		Precision	Recall	F1	Precision	Recall	F1
Review+Profile+Image	Taiyi-CLIP		N/A	95.42	95.38	95.40	93.21	93.72	93.46
Review+Profile+Image	Qwen-VL		N/A	96.05	96.01	96.01	94.31	93.99	94.11
Review	RoBERTa	N/A	N/A	96.16	96.15	96.15	94.18	94.58	94.37
Review+Image	BERT	ViT	Combine	96.88	96.85	96.86	95.37	95.77	95.56
Review+Profile+Image	RoBERTa	ViT	Combine	97.23	97.22	97.22	96.03	96.11	96.06

Presented results are on the test dataset.

Table 5
Best experimental settings for fake review detection.

Parameter	Best setting
Text encoder	RoBERTa
Image encoder	Vision Transformer (ViT)
Fusion strategy	Combined Fusion
Batch size	16
Initial learning rate	5e-05
Number of epochs	15
Optimizer	Adam
Performance (F1)	Macro F1: 0.96, Avg F1: 0.97

5.4. Misclassification analysis

This experiment visualized the classification results of the top-performing model in identifying various false review intentions, as depicted in Fig. 5.

- **Real Reviews:** The accuracy was high at 99.22%, with a recall rate of 99.05%. Of 3754 real reviews processed, misclassifications were minor, including 10 as IE, 9 as CD, and 11 as IC. Out of 3574 false reviews, 28 were erroneously classified as genuine, mainly involving 10 as IE.
- **Incentivized Exaggeration (IE):** The model reached an accuracy of 92.74% and a recall of 93.98%. From 897 reviews falsely labeled as IE, 29 were misidentified as TA and 14 as IC. Of the 909 identified as IE, 54 were incorrectly classified, largely as TA.
- **Competitive Denigration (CD):** Demonstrated an accuracy of 98.35% and a recall of 98.78%. Misclassifications included only 11 from 903 reviews tagged as CD. Additionally, from 907 identified as TA, 15 were mistakenly classified as CD.
- **Traffic Attraction (TA):** Achieved an accuracy of 93.78% and a recall of 95.16%. From 888 reviews erroneously labeled as TA, 32 were incorrectly judged as IE, and from those identified as TA, 56 were of different types, mainly IE and IC.
- **Irrelevant Commentary (IC):** Attained an accuracy of 96.06% and a recall of 93.57%, with misjudgments including 23 instances classified as IE and 21 as TA. Among those identified as IC, 34 were from different categories, primarily real reviews and IE.

The model exhibited robust accuracy in identifying fake review intentions, with particularly high performance for real reviews (accuracy: 99.22%, recall: 99.05%). Misclassifications were most frequent between similar types of fake intentions, notably between Incentivized Exaggeration (IE) and Traffic Attraction (TA). Competitive Denigration (CD) and Irrelevant Commentary (IC) also showed high accuracy but experienced occasional misclassifications. Overall, the model effectively distinguished between real and fake reviews, though some challenges remained in accurately categorizing all types of fake review intentions.

5.5. Statistical performance comparisons of model configurations

5.5.1. Comparison between inputs

To validate the robustness of our framework, we conducted paired t-tests to evaluate performance differences across models with various

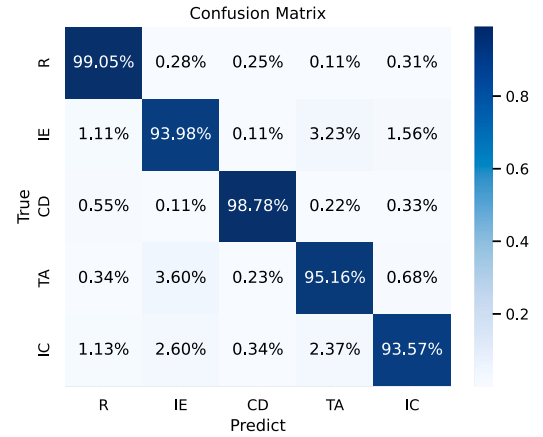


Fig. 5. Confusion matrix.

inputs, using Macro F1 scores as the metric. As shown in Table 6, for the first comparison between the “review + image” and “review + image + profile” configurations, we observed a significant mean decrease in performance of -0.505 , with a standard deviation of 0.247 and a standard error of 0.055 . The 95% confidence interval was from -0.621 to -0.390 , yielding a significant t-statistic of -9.145 and a p -value of $2.17\text{E}-08$. In the second comparison, “review” versus “review + image + profile”, there was a larger mean decrease of -1.066 , with a standard deviation of 0.342 and an SEM of 0.076 . The confidence interval was notably wide at -1.226 to -1.223 , with a t-statistic of -13.945 and a highly significant p -value of $1.97\text{E}-11$. These findings strongly indicate that incorporating image and profile information significantly boosts performance compared to using only review text or review text with images.

5.5.2. Comparison between encoders

We further examined performance variations between different encoder and image-processing models using paired t-tests. As shown in Table 7, the comparison between the Chinese-BERT and Chinese-RoBERTa encoders showed a mean performance difference of -0.090 , with a standard deviation of 0.202 and a standard error of 0.064 . The 95% confidence interval ranged from -0.234 to 0.055 , and the t-statistic was -1.407 , indicating no statistically significant difference (p -value = 0.193).

Conversely, the comparison between the ResNet and ViT (Vision Transformer) image-processing models revealed a more substantial mean decrease of -0.559 in performance when using ResNet. The standard deviation was 0.268 , and the SEM was 0.085 . The confidence interval spanned from -0.751 to -0.367 , with a t-statistic of -6.592 , demonstrating a highly significant difference (p -value = $1.00\text{E}-4$). This analysis confirms that ViT significantly outperforms ResNet, emphasizing the critical role of selecting the optimal encoder to enhance model efficacy.

Table 6
Pair t-test for different inputs.

		Difference between pairs					t-Stat	DF	Sig. (2-tailed)
		Mean Diff.	Std Dev	SEM	95% CI Lower	95% CI Upper			
Pair 1	Review + image – Review + image + profile	–0.505	0.247	0.055	–0.621	–0.390	–9.145	19	2.17E–08
Pair 2	Review – Review + image + profile	–1.066	0.342	0.076	–1.226	–1.223	–13.945	19	1.97E–11

Table 7
Pair t-test for different encoders.

		Difference between pairs					t-Stat	DF	Sig. (2-tailed)
		Mean Diff.	Std Dev	SEM	95% CI Lower	95% CI Upper			
Pair 1	Chinese–BERT – Chinese– RoBERTa	–0.090	0.202	0.064	–0.234	0.055	–1.407	9	0.193
Pair 2	ResNet – ViT	–0.559	0.268	0.085	–0.751	–0.367	–6.592	9	1.00E–4

Table 8
ANOVA test for different fusion strategies.

	Sum of squares	df	Mean square	F	Sig.
Between groups	0.226	4	0.057	0.423	0.790
Within groups	2.004	15	0.134		
Total	2.231	19			

5.5.3. Comparison between fusion strategies

The efficacy of various fusion techniques was assessed using an ANOVA test to explore performance variance across different fusion types. As shown in Table 8, the analysis revealed a total sum of squares of 2.231 and a between-group sum of squares of 0.226 over four degrees of freedom, resulting in a mean square between groups of 0.057. The calculated F-statistic was 0.423, corresponding to a p -value of 0.790. These results indicate that there is no statistically significant difference in performance among the fusion strategies, suggesting that the choice of fusion technique may not critically affect model performance in this context.

This finding implies that while image encoder selection and input configuration significantly influence model performance, the specific type of fusion technique employed does not lead to substantial performance variations.

6. Discussion and implications

6.1. Discussion of key findings

To address the gap in fake review detection, this study introduces a novel task Fake Review Intention Detection in Review Context (FRIDRC), constructs a new dataset, develops a multimodal framework, and conducts extensive experiments. Our key findings are:

1. The integration of multiple modalities — specifically, combining review text with multimodal review context (images and profiles) information — improves the model's ability to detect fake review intentions. The use of review text combined with images and profiles yielded the highest precision, recall, and F1 scores, demonstrating the effectiveness of review context.

2. The misclassification analysis demonstrated exceptional accuracy for real reviews (99.22%) and managed to effectively distinguish between different fake review intentions. However, challenges persisted in accurately differentiating similar categories of fake reviews, particularly between Incentivized Exaggeration and Traffic Attraction.
3. Statistical analyses confirmed the superiority of using a combination of review text, images, and profile information over configurations with fewer modalities. Moreover, the Vision Transformer (ViT) outperformed ResNet, highlighting the critical role of encoder selection in model performance. Nonetheless, the type of fusion technique and text encoders employed did not significantly affect the outcomes.

6.2. Theoretical implications

Theoretically, this study not only pioneers the FRIDRC task but also demonstrates the efficacy of our multimodal framework, laying the groundwork for future research in fake review detection. Our findings challenge the traditional binary classification of reviews (Salminen et al., 2022) by introducing a more nuanced categorization based on varying intentions. We identify four distinct types of fake review intentions and curate a dataset that includes both human-generated and AI-generated samples to support their detection.

Moreover, the inclusion of multimodal context information has proven its superiority in predictive tasks concerning online reviews (Liu et al., 2021; Nguyen et al., 2023; Wu et al., 2024). Our findings align with and extend current theories by demonstrating how the integration of review contexts can improve the inference capabilities of machine learning models in detecting fake reviews. This advancement supports the theoretical assertion that contextual richness is crucial in complex detection tasks like intention analysis, moving beyond traditional text-centric methods.

6.3. Practical implications

The practical applications of our research are extensive, especially for online platforms and businesses reliant on user reviews. Our tool not only assesses the authenticity of reviews but also discerns their underlying intentions. This dual capability improves the ability to monitor and detect fake reviews, whether generated by humans or AI. Implementing

our framework could bolster the efforts of businesses to safeguard their brand reputation and provide consumers with trustworthy information, thus enhancing consumer trust and business performance (Munzel, 2016).

Additionally, the evidence provided by our study on the effectiveness of modalities and specific representation and fusion modules can guide businesses and developers in customizing their AI systems to be more robust against sophisticated fake review tactics.

Furthermore, a deeper understanding of the intentions behind fake reviews can aid in developing more effective policies to protect consumers and businesses from the adverse impacts of misleading content (Zaman et al., 2023). Policymakers can leverage the insights from our study to craft more precise guidelines and regulations concerning online review management.

6.4. Limitations and future work

This study has several limitations: the dataset is relatively small and linguistically homogeneous, which may hinder the generalizability of the model to diverse platforms and languages. Moreover, the framework primarily refines existing modules from other fields for fake review detection. While effective, this approach does not introduce fundamentally novel methods. Additionally, the use of single-frame images from video key frames may not adequately represent comprehensive video features, potentially limiting the effectiveness of multimodal analysis. Finally, the high recognition accuracy using only review text suggests a low complexity in the dataset, which might not fully challenge the model against more sophisticated or subtle fake reviews found in practical applications.

Future research should aim to diversify the dataset with more languages and media types to enhance the model's generalizability. Developing novel methodologies tailored to fake review detection could further refine accuracy and interpretability. Incorporating comprehensive video analysis could better capture multimedia features, improving detection capabilities. Additionally, creating a more sophisticated dataset that mimics real-world deceptive tactics could provide a rigorous challenge to the model. Integrating adaptive learning mechanisms would also help the model stay effective against evolving fake review strategies.

7. Conclusion

This study addresses the growing challenge of fake reviews by introducing the novel task of FRIDRC. To explore the detection capabilities for this task, we developed a multimodal dataset that encompasses various fake review intentions and implemented a multimodal framework with decoupled encoders and diverse fusion strategies. Based on the experiments, the best-performing model was the combination of RoBERTa for text encoding and the Vision Transformer (ViT) for image encoding, using a combined fusion strategy. This configuration significantly outperformed other tested models including advanced pre-trained models, achieving a macro F1 score of 0.96 and an average F1 score of 0.97.

These results reveal that this framework achieved superior performance in detecting fake review intentions. Furthermore, the findings underscore the importance of incorporating multimodal review context information to improve the efficacy of fake review detection in diverse contexts.

CRedit authorship contribution statement

Jingrui Hou: Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Zhihang Tan:** Writing – original draft, Visualization, Formal analysis, Data curation. **Shitou Zhang:** Validation, Software, Methodology, Conceptualization. **Qibiao Hu:** Writing – review & editing, Supervision, Resources. **Ping Wang:** Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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